

The Link Wars Debate: Proprietary, Open and Atomic Fabrics in AI Networking

September 8, 2025



Format: Lively head-to-head debate with 4-6 panelists + audience.

Interconnect Landscape

- **NVLink:** NVIDIA's proprietary GPU-centric link fabric supporting shared memory and collective offload within scale-up clusters. Ideal for ultra-low-latency, high-bandwidth workloads.
- **UALink:** An industry consortium (AMD, Intel, Meta, Google, Cisco, HPE, AWS et al.) open standard for scale-up links (200 Gbps/lane, up to 800 Gbps), offering memory semantics, lower power/latency vs PCIe and vendor-neutral scale-up fabric.
- **Ultra-Ethernet (UEC):** Open Ethernet-based scale-out fabric tailored for AI/HPC with in-network flow control, congestion management, ultra-low tail latency, and support up to terabit speeds across racks.
- **AELink (Æthernet):** A deep reinvention of Ethernet spearheaded by OCP. Æthernet exposes ACID-style transactional behavior at the NIC and link layer—atomicity, isolation, durability and consistency—rejecting “best-effort” networks for reliable, deterministic communication.

Why AELink (Æthernet) Matters

- **Network-as-transaction:** Unlike traditional fabrics, Æthernet treats messages as secure, ACID-like transactions visible only between endpoints—no half-delivered frames or hidden drops.
- **Programmability at the edge:** Enables app-specific logic and reversible links directly at NICs—moving function into the network interface and enabling host bypass and in-network compute playbooks.
- **Long-term vision:** Designed not just for current AI clusters, but for the next 50 years—chiplet meshes, reversible links, quantum-aware, truly application-aware fabrics.

Why This Session is a Must-Attend

- **Topic magnetism:** $\mathcal{A}$ Ethernet brings cutting-edge theory, new models of compute-linked networking, and standardization toward long-term fabric correctness.
- **Engagement:** Debate infuses drama—proprietary vs open, scale-up vs scale-out vs transactional semantics—and challenges attendees to rethink Ethernet itself.
- **Actionable insight:** Attendees will understand real trade-offs: latency, power, programmability, lock-in, and future architectures.

Panel Proposal and Structure

Panelists (example lineup):

- NVLink champion (e.g., NVIDIA or systems integrator)
- UALink consortium representative (AMD, Meta, Cisco)
- Ultra-Ethernet advocate (UEC, Synopsys)
- AELink project lead (OCP OAE / e.g., David Johnston)
- PCIe/CXL. consortium representative (Intel, SNIA)
- Fibrechannel. Industry expert. Spec writer?

Structure:

1. Opening positions: NVLink, UALink, AELink, PCIe//CXL & UEC
2. Head-to-head challenges:
 - “Can open UALink pods replace NVLink clusters?”
 - “Will Ultra-Ethernet displace InfiniBand and RoCE?”
 - “Does $\mathcal{A}$ Ethernet redefine what ‘network’ means?”
3. Audience Q&A: on Incremental vs. Revolutionary innovation.

Current Title: NETC-304-1: Networks and Links: Advanced Topics

New Abstract: *The Link Wars Debate: Proprietary, Open and Atomic Fabrics in AI Networking.* AI-scale infrastructure is moving past traditional switches and routers. The future lies in **high-speed links**—NVLink, UALink, AELink, and Ultra-Ethernet – delivering memory semantics, programmability, and deterministic messaging. Join this **high-energy debate** as experts square off on proprietary vs open vs atomic link fabrics, debating which model should power next-generation AI systems—and why it matters for scalability, correctness, & vendor choice.